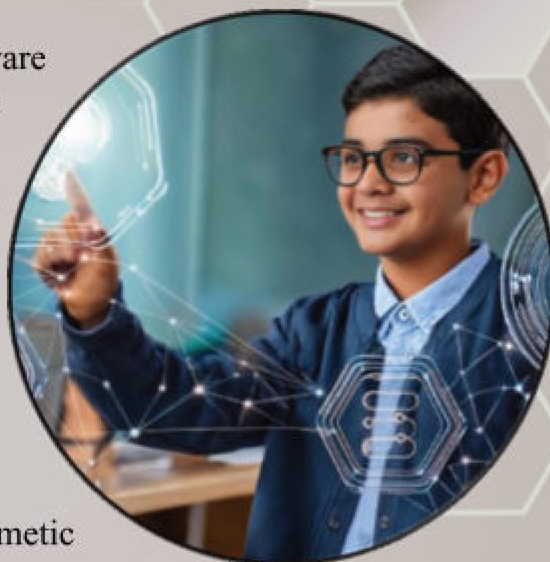
Chapter 2

Digital Skills

Student Learning Outcomes (SLOs):

By the end of this unit, students will be able to:

- Basics of Worksheets
- Purpose and uses of spreadsheets software
- Different spreadsheet software e.g. Excel, Google Sheets, Open Office
- Creating result card, home budgets, timetables etc. using spread sheets.
- Preparing Worksheet
- Organize data in worksheet within a workbook
- Select a range of cells
- Add borders
- Use simple built-in functions
- Create simple formulae (arithmetic operations)
- Create an appropriate chart for data presentation.



2.1 Introduction:

In today's society, computer skills are as essential as reading and writing. They are essential for everyday technological use, education, and future employment prospects. Learning digital abilities enables you to communicate, solve issues, and create using technology.

This chapter focusses on digital skills and how to organize data and create simple charts using spreadsheet programs like Google Sheets and Microsoft Excel.

2.2 Spreadsheet:

A spreadsheet is a digital tool used to organize, calculate, and analyze data in rows and columns (tabular form). It looks like a grid where you can enter information, such as numbers, text, or formulas, to perform calculations and make charts.

Excel Spreadsheet Figure 2.1

The most common spreadsheet programs are Microsoft Excel, Google Sheets and Open Office.

Microsoft Excel	Open Office	Google Sheet
Microsoft Excel is a powerful spreadsheet application developed by Microsoft and is a component of the Microsoft Office productivity package. Excel is widely used in business, finance, accounting, data analysis, and personal productivity due to its versatility and large feature set.	OpenOffice, also known as Apache OpenOffice, is a free open-source software suite that includes tools for word processing, spreadsheets, presentations, graphics, and databases. It is a popular choice for individuals, organizations, and schools seeking an affordable alternative to commercial office software.	Google Sheets is a free online application for creating, editing, and storing spreadsheets. It is part of Google Drive and is extensively used for personal, professional, and educational projects because it is simple to use, allows people to share information in real time, and integrates with other Google services.

2.2.1 Purpose of Spreadsheet:

A spreadsheet is a tool used to organize, calculate, and manage information. It helps you work with lots of data in an easy and clear way. A spreadsheet is like a digital version of a paper worksheet. A spreadsheet can be useful for

- **Organizing Data:** neatly into rows (side to side) and column (up and down) making it easy to read and find what you need.

- **Doing Calculation:** working with mathematical formulas (add, subtract, average number etc.) and performing more complex calculations.

Fig 2.2 Purpose of Spreadsheet

- **Analyzing Data:** to look at data closely to see patterns or trends. You can sort, filter and summarize data to find important information.

- **Creating Charts and Graphs:** Data turns into charts and graphs using spreadsheet, making it easier to understand and share with others.

Fig 2.3 Working with Spreadsheets

2.3 Microsoft Excel:

Microsoft Excel is a computer program used for creating and working with spreadsheets. It is part of the Microsoft Office suite which provides tabular sheets (*having cells that contains data in rows and column*) that makes it easier to analyze, organize and managing data. It is popular in businesses to perform financial analysis.

Following are the common uses of Microsoft Excel:

Personal Use: Track your budget, manage a list of contact, maintain your monthly expense

School: Organize homework assignments, track grades and create graphical charts.

Work: Analyze sales data, manage employee schedules or track the financial result of companies.

2.4 Working with Microsoft Excel:

- **Start Microsoft Excel:** Click on windows icon and type **"Excel"** in search box. Excel app icon will visible click on it or click open option showing as in below figure.

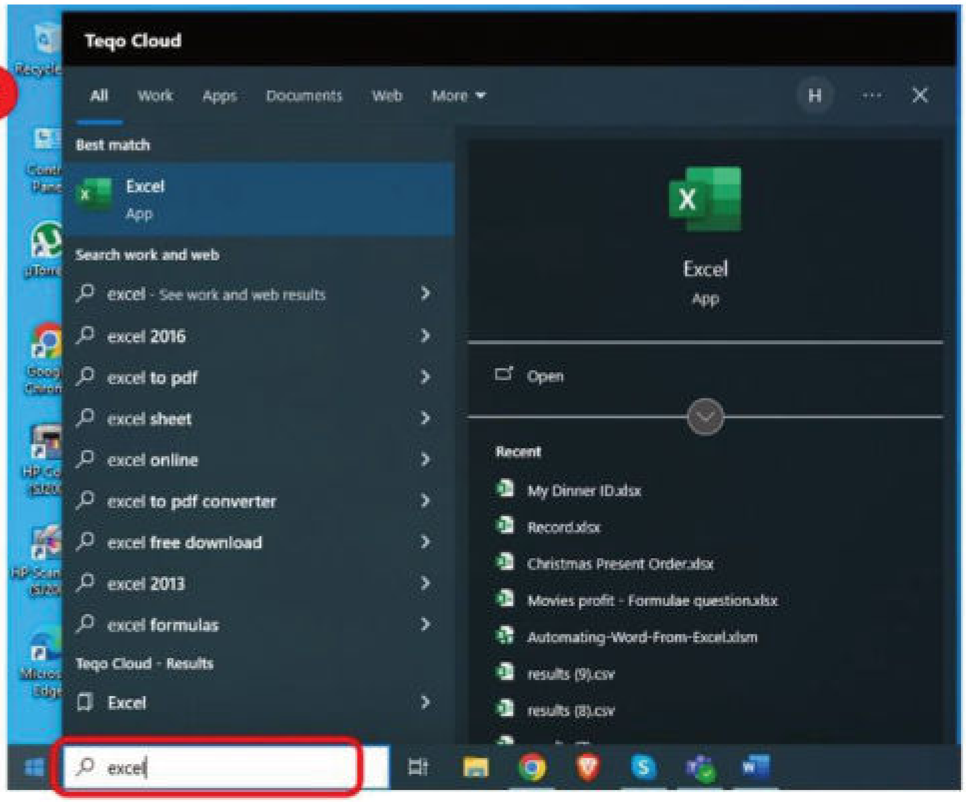
Start Microsoft Excel

❶ Click on windows icon start button.

❷ Type "Excel" in search text box.

❸ Click on Excel icon to start it.

2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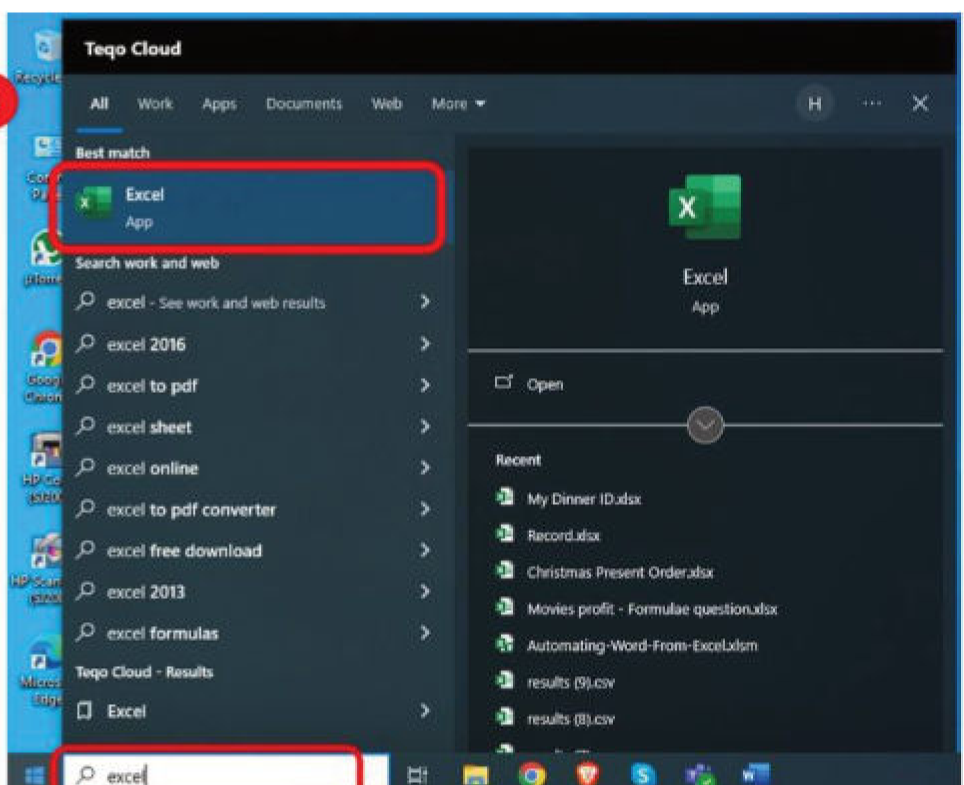
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excel

- **Create a New Workbook:** When Excel application is open, you will be given the option to start with a blank workbook or choose a template. A blank workbook is the best place to start.

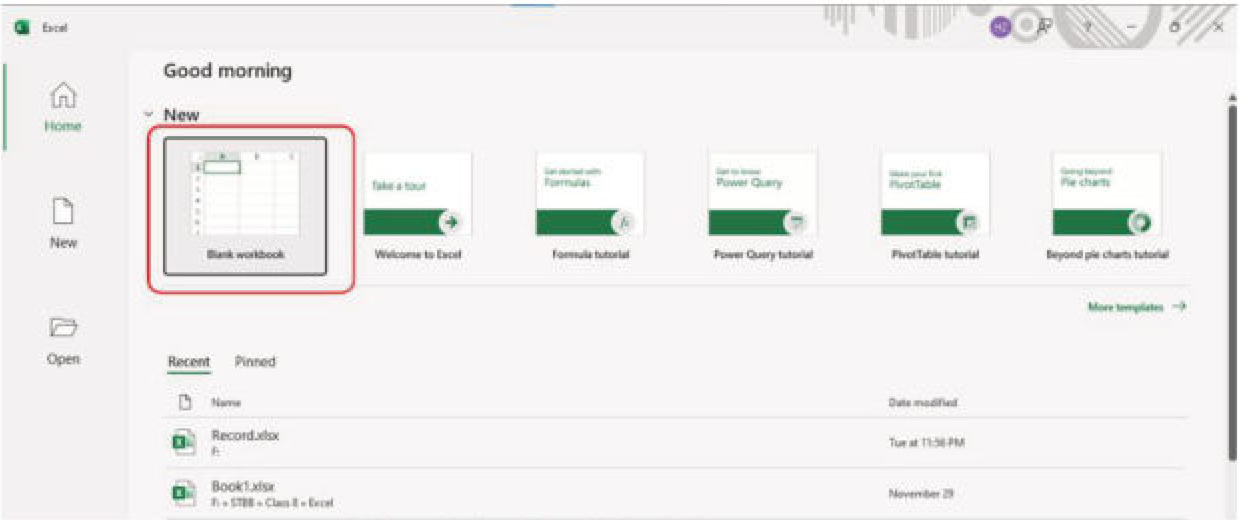


Figure 2.4(a) Excel Blank document open

- **Opening an Existing Workbook:** If you want to open an existing file, click **File > Open** and choose the file you want. There are some more options under Open which are as fellow
  - **Recent:** it will show you the last saved worksheet as showing in below figure.

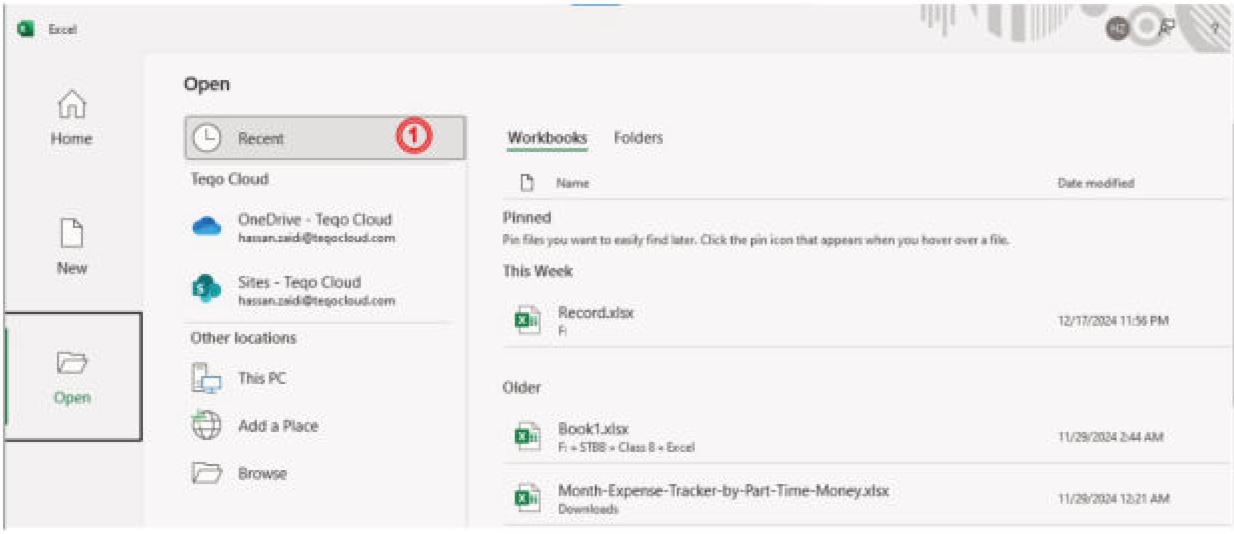


Figure 2.4(b) Excel Open Recent documents

- **This PC:** it will show you the folders present in My Document where you can open your last saved worksheet.

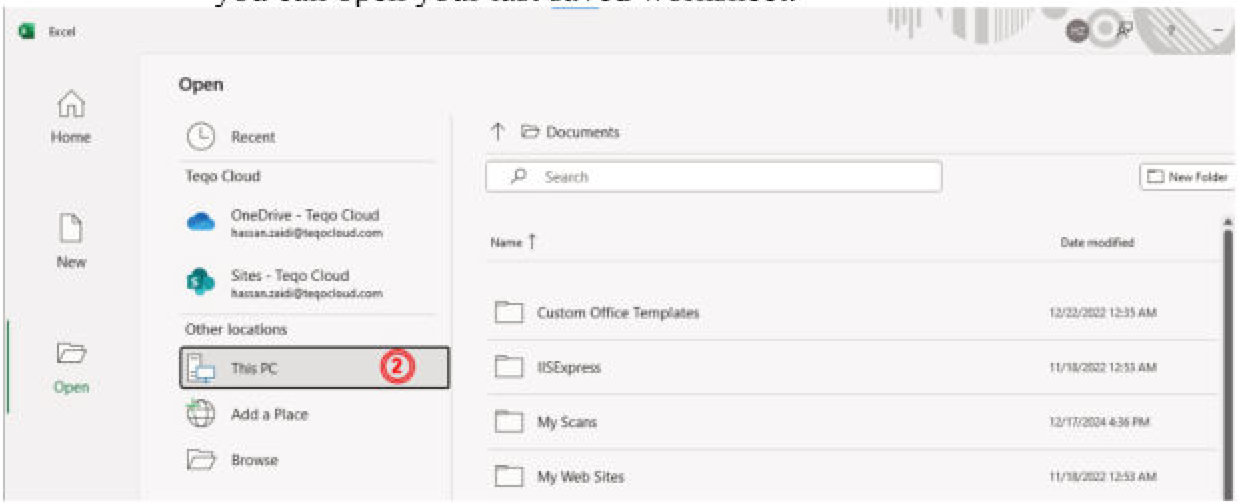


Figure 2.4(c) Excel Open documents from My PC

- **Browse:** it will show a dialog box from which you can select any drive from your computer and go to you desired folder to open your existing excel file.

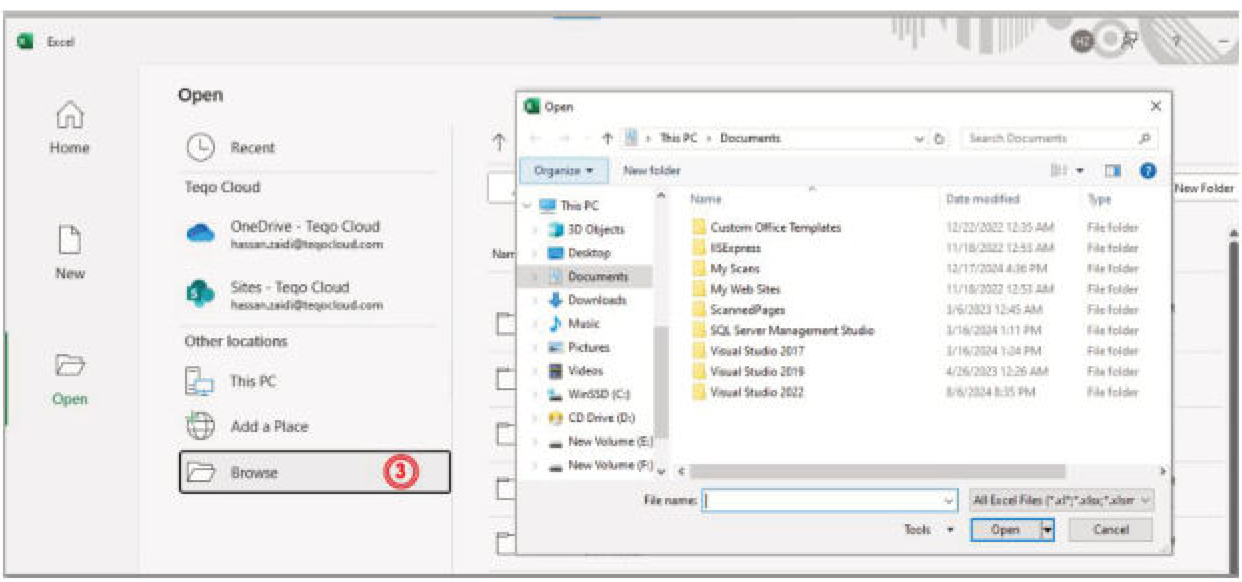


Figure 2.4(d) Excel opening browse window

2.4.1 Microsoft Excel Interface (Version 2021):

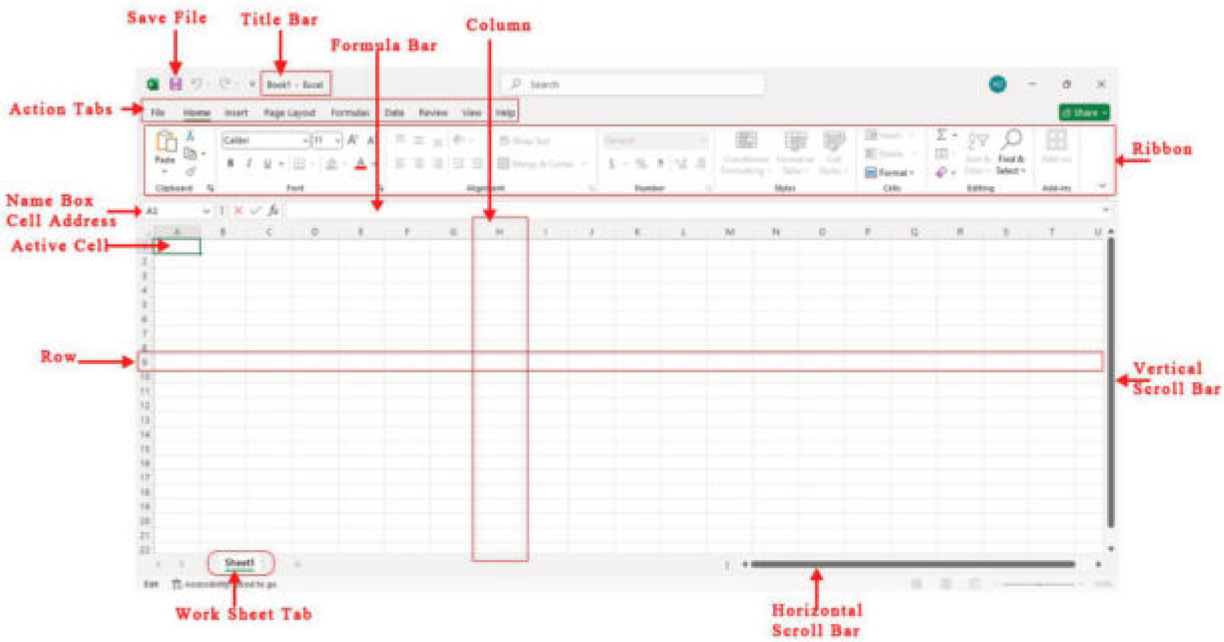


Fig 2.5 Microsoft Excel Interface

Workbook:

A workbook represents an Excel file containing one or more worksheets. It functions similarly to a binder, holding all of the sheets, charts, and data. Each workbook can hold several worksheets, and you can save them using the extension.xlsx.

Worksheet:

A worksheet is a single page or tab of a workbook. It consists of a grid of rows and columns in which you may enter and manipulate data. A workbook may contain several worksheets, each worksheet is a grid of rows and columns where you can enter data. The rows are numbered (1, 2, 3,...) and the columns are labeled with letters (A, B, C,...). Worksheet represented by a tab at the bottom of the screen. Each worksheet contains its own set of data and can be worked on independently.

Cells:

Each box in the grid is called a cell. Each cell has a unique reference, such as A1 (column A, row 1) or B2 (column B, row 2).

Ribbon:

The top section of the Excel window is called the Ribbon, where you'll find different tabs like Home, Insert, Formulas, Data, Review and more. Each tab contains tools and options for working with your data.

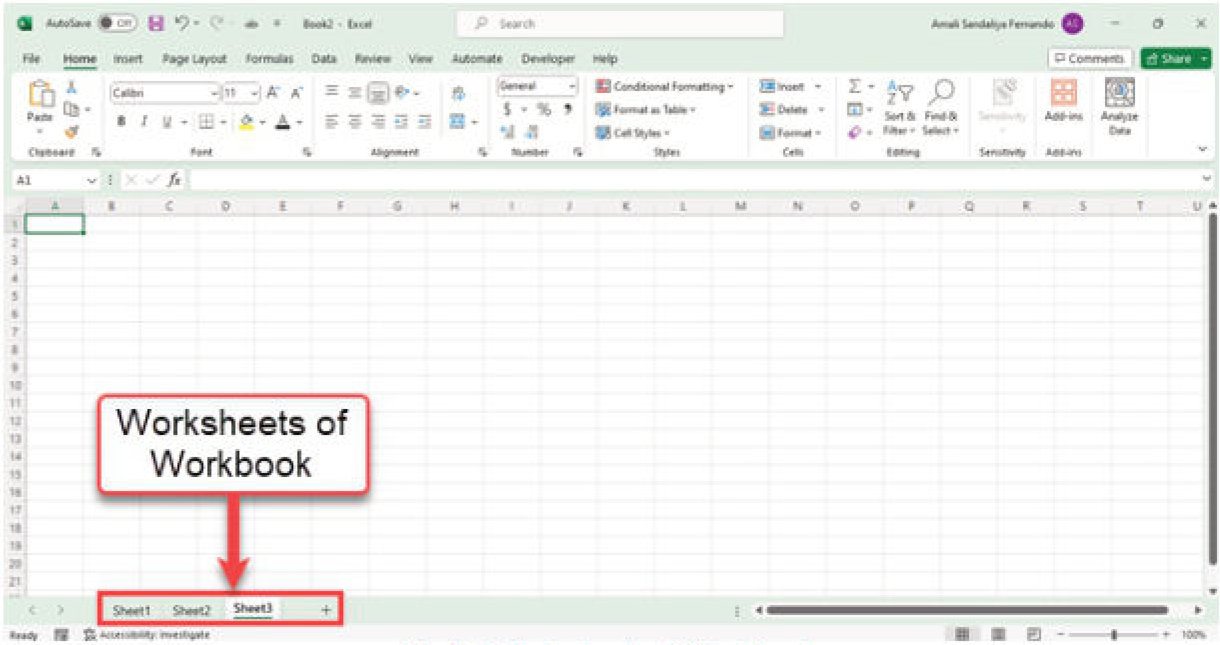


Fig 2.6 Worksheets of Workbook

Quick Tip:

Tabs: (e.g., "Home", "Insert", “Draw”, "Layout") to view the commands in that tab.
Groups: Within each tab, tools are grouped by category. For example, in the "Home" tab, the Font group contains tools for text formatting, and the Editing group has tools for finding, selecting, and clearing data.
Commands: Click on any command (e.g., Bold, Insert Table, or AutoSum) to perform the action.

2.4.2 Home Tab:

The Home tab provides basic options to perform tasks like formatting text, organizing data, performing calculations, and managing how your information is displayed.

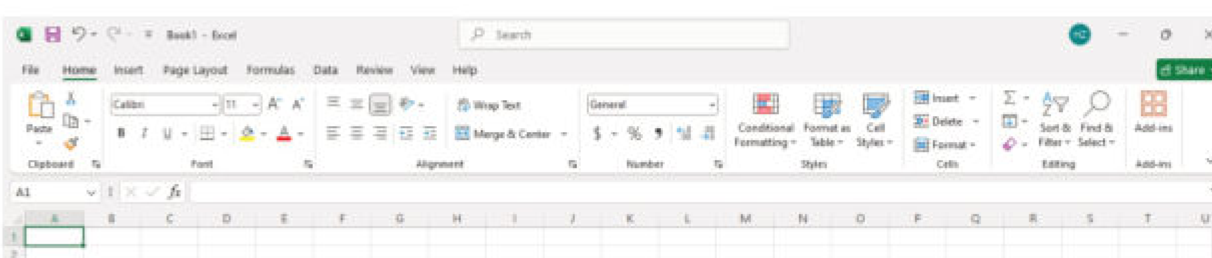


Fig 2.7 Home Tab options

Quick overview of Home Tab Group

Group	Key Tool
Clipboard	Cut, Copy, Paste, Format Painter
Font	Font Style, Size, Bold, Italic, Underline, Color, Borders
Alignment	Left/Center/Right Align, Merge, Wrap Text, Orientation
Number	Number Format, Increase/Decrease Decimal, Currency, Percent
Style	Conditional Formatting, Format as Table, Cell Styles
Editing	Find & Select, Clear, Sort & Filter, AutoSum

Home tab provides following features:

- **Clipboard Group:**

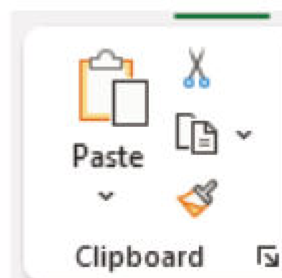


Fig 2.7(a)
Clipboard Options

Cut: Removes the selected content and put it on the clipboard so you can paste it somewhere else. Short key: **Ctrl + X.**

Copy: Makes a copy of the selected content on the clipboard without removing it from its original location. Short Key: **Ctrl + C.**

Paste **Paste:** is used to pastes the content you have copied or cut. Short Key: **Ctrl + V**

Format Painter: Copies the formatting (colors, fonts, borders) of a cell and applies it to another cell. Short Key: Use **Alt + Ctrl + C** to copy a format, and **Alt + Ctrl + V** to paste a format.

- **Font Group:**

Font Style: Change the font of your text (e.g., Arial, Calibri). Short Key: **Ctrl + Shift + F**
Font Size: Change the size of the text. Bold, Italic, Underline: Make the text bold, italic, or underline. Short Key: **Ctrl + Shift + P**
Font Color: Change the color of the text.

Fig 2.7(b) Font Group options

- **Alignment Group:**

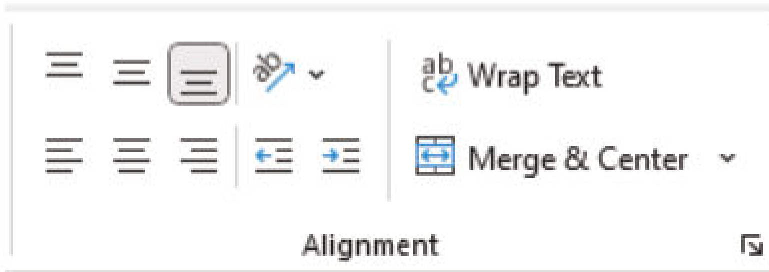


Fig 2.7(c) Alignment Group options

Align Text Left, Center, Right: Align the text in the selected cells to the left, center, or right.

Merge & Center: Combines multiple selected cells into one and centers the text within the merged cell.

Wrap Text: Allows text to wrap within the cell, so it doesn't overflow into adjacent cells.

Orientation: Change the angle of the text (e.g., rotate the text to make it vertical or at an angle).

top, **middle**, and **bottom** alignment options are used to position the content of a cell vertically. Click on any of these icons to adjust the alignment of the content in the selected cells.
Short Key: **Alt** + **H** + **A** + **T** (Top Align), **Alt** + **H** + **A** + **M** (Middle Align), **Alt** + **H** + **A** + **B** (Bottom Align)

- **Number Group:**

  Change the number format (e.g. General, Currency, Percentage, Date etc.). It customizes how the data displayed in cell.

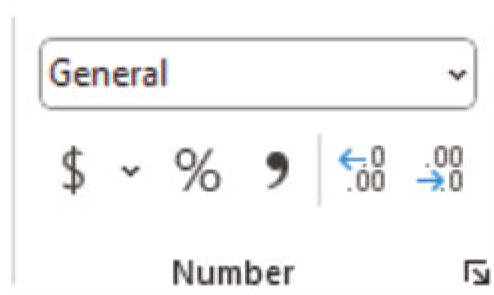


Fig 2.7(d) Number Group options

Apply number formats such as dates, currency, or fractions to cells in a worksheet. For example, if you are working on your quarterly budget, you can use the Currency number format so your numbers represent money. Or, if you have a column of dates, you can specify that you want the dates to appear as March 14, 2012, 14-Mar-12, or 3/14.

Follow these steps to format numbers:

1. Select the cells containing the numbers you need to format.

2. Select CTRL+1.
3. In the window that displays, select the Number tab.
4. Select a Category option, and then select specific formatting changes on the right.

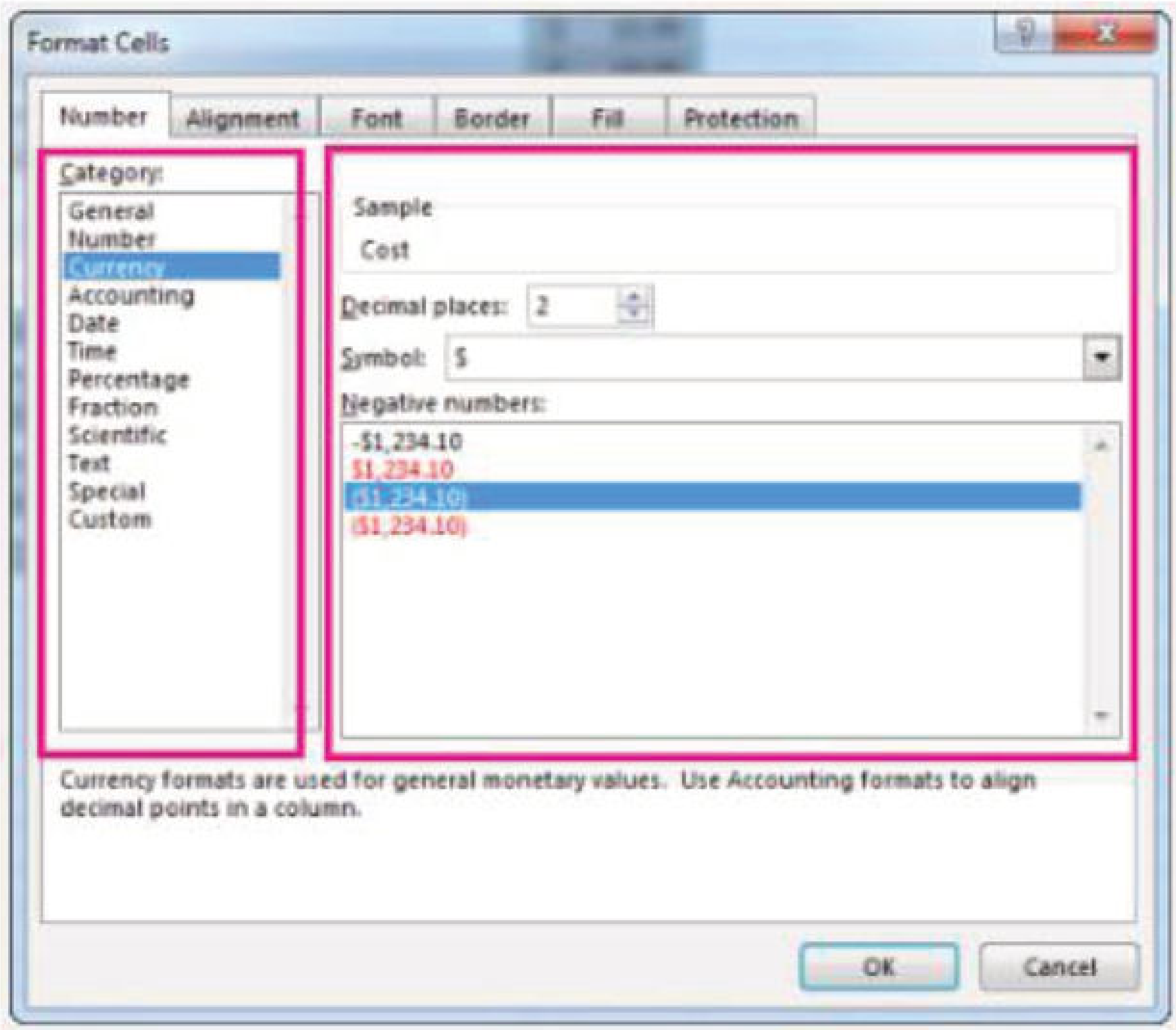


Fig 2.7(e) Style Group options

- **Style Group:**
  Conditional Formatting: Highlights cell based on certain rules (e.g. values above a threshold).
  Cell Style: Apply predefined styles to cells for quick formatting.

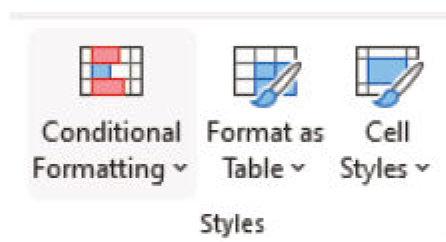


Fig 2.7(f) Style Group options

2.4.2.1 Creating a Sales Report:

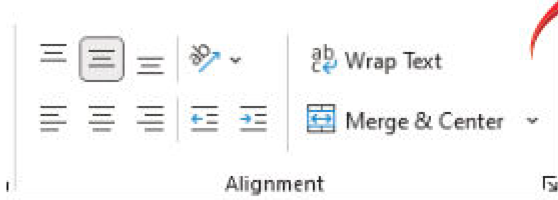
We are going to format and organize data for a simple sales report using options from the Home Tab by performing following steps.

1. Open Excel software and create enter following data.

	A	B	C	D
1	Product	Quantity	Price	Total
2	Apple	10	2.5	
3	Banana	15	1.8	
4	Orange	8	3	
5				
6				
7				

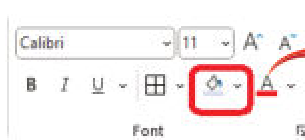
2. Apply basic formatting by using font group:
   - Select the table header (Product, Quantity, Price, Total) and press "**B**" from Font group to bold the selected text.

   - Center aligns the headers using Alignment Group.



Teacher demonstrates the students how to apply different alignment on text.

   - Add background color to the header (e.g. light blue) using **Fill Color**.



Teacher demonstrates the students how to apply different fonts, background and text color.

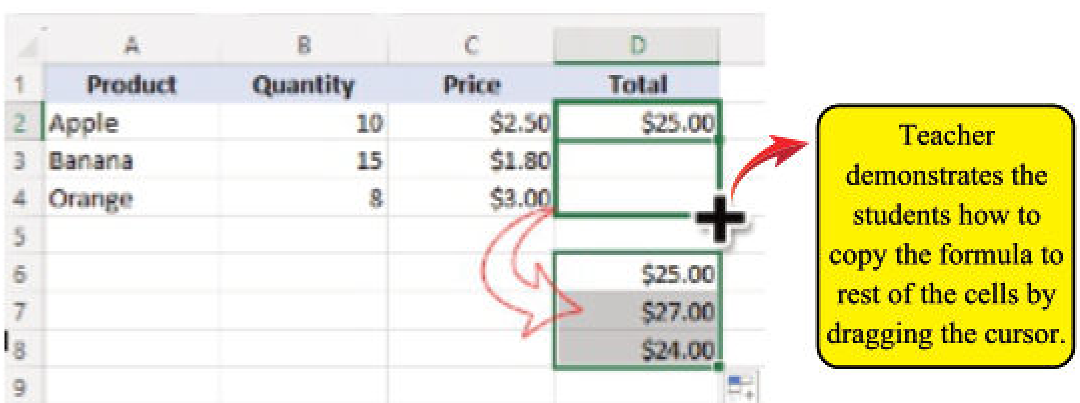
- Following image is showing the above steps result.

	A	B	C	D
1	Product	Quantity	Price	Total
2	Apple	10	2.5	
3	Banana	15	1.8	
4	Orange	8	3	
5				

3. Calculate Totals:
   In column D (Total) we are using a formula (***will discuss in this Chapter 2.5***) to calculate the total for each product (Quantity * Price).
   Example: In cell D2, enter =B2 *C2 as showing in below picture.

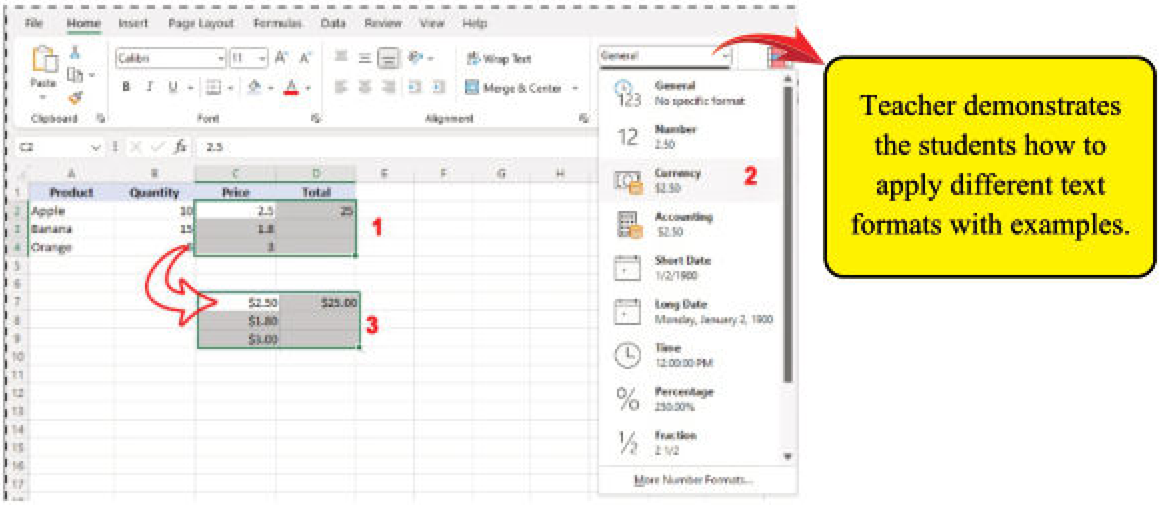
	A	B	C	D
1	Product	Quantity	Price	Total
2	Apple	10	2.5	=B2*C2
3	Banana	15	1.8	
4	Orange	8	3	

After entering the formula select the cell and move your cursor to the bottom right of the selected cell now your cursor change to + sign drag up to the cells where you want the calculation as showing in below picture.

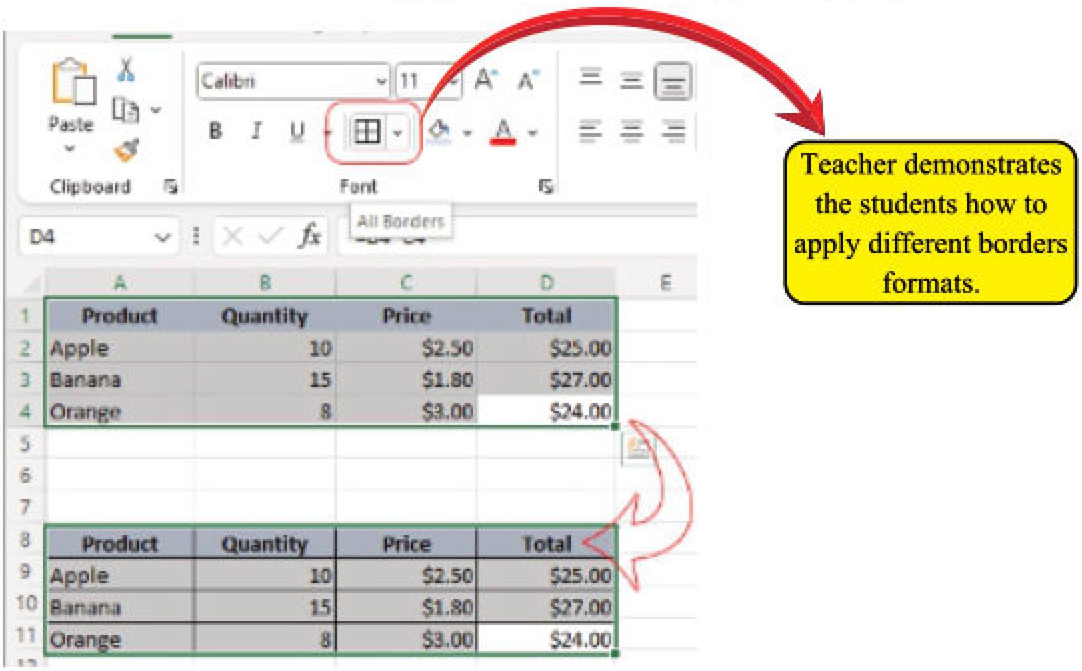


	A	B	C	D
1	Product	Quantity	Price	Total
2	Apple	10	$2.50	$25.00
3	Banana	15	$1.80	
4	Orange	8	$3.00	
5				
6				$25.00
7				$27.00
8				$24.00
9				

4. Apply Number formatting:
   Format the Price and Total columns to currency format using the **Number** group.



5. Apply Borders:
   Select the entire table and apply **All Borders** using Font group.



6. Save Your work:
   Save the file with the name BasicActivity.xlsx

2.4.3 Insert Tab:

The Insert tab in Microsoft Excel is a powerful tool use to add various objects, charts, and features to your worksheet. It contains all the tools you need to insert elements like tables, pictures, charts, shapes, and more in order to make your Excel workbook more interactive and informative.

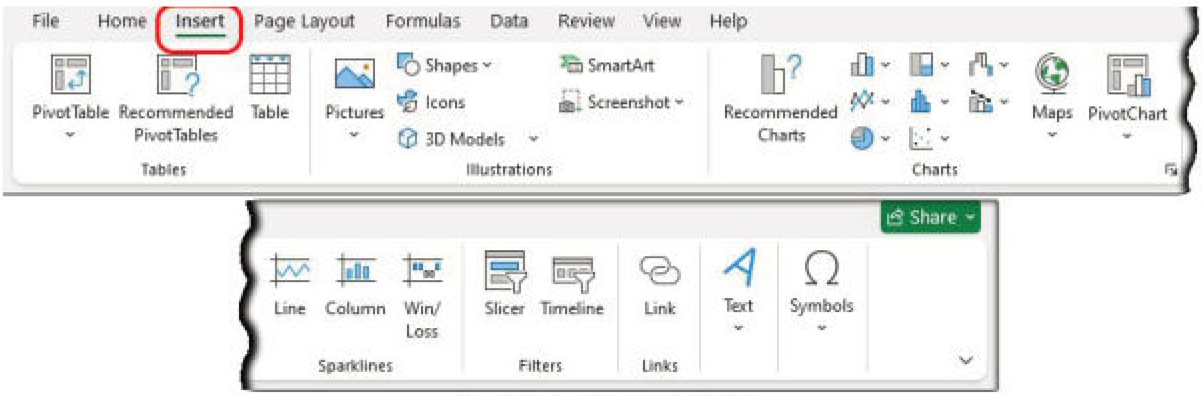


Fig 2.8 Insert Tab options

2.4.4 Page layout Tab:

The Page Layout tab in Microsoft Excel is where you can control the appearance and formatting of your worksheet for printing. It allows you to adjust page orientation, paper size, margins, and themes, as well as fine-tune how data will fit on the printed page. By using the tools in this tab, you can ensure that your Excel worksheet appears professional and is properly formatted for printing or sharing with others.

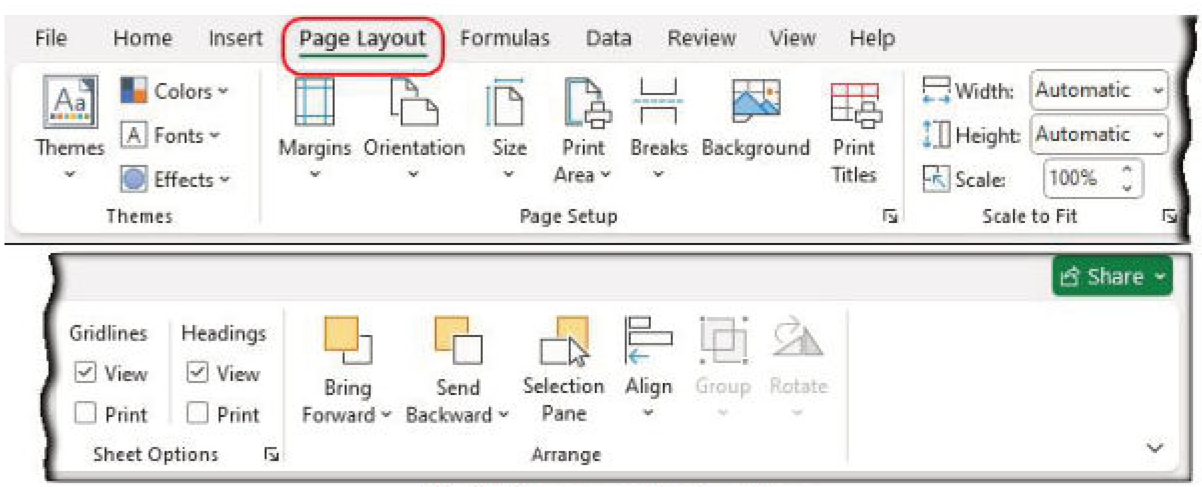


Fig 2.9 Page layout tab options

2.4.5 Formulas Tab:

The Formulas tab in Microsoft Excel provides the tools and functions related to creating and managing formulas in your workbook. It provides access to a wide range of built-in functions for calculations, as well as tools to help you organize and test your formulas. This tab is essential for anyone working with numerical data or performing complex calculations in Excel.

- Insert function helps to insert different formulas by providing their parameters.
- Auto Sum perform quick addition calculation on your worksheet.

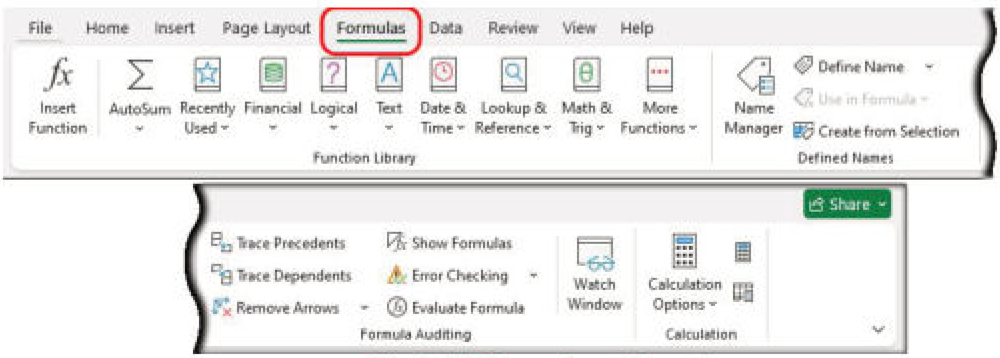


Fig 2.10 Formulas tab options

2.4.6 Data Tab:

The Data tab in Microsoft Excel provides tools for organizing, sorting, filtering, importing, and transforming your data into your desired format.

- Sort option helps to arrange data in specific order (Ascending or Descending)
- Filter option allows to easily extract the records based on some criteria.

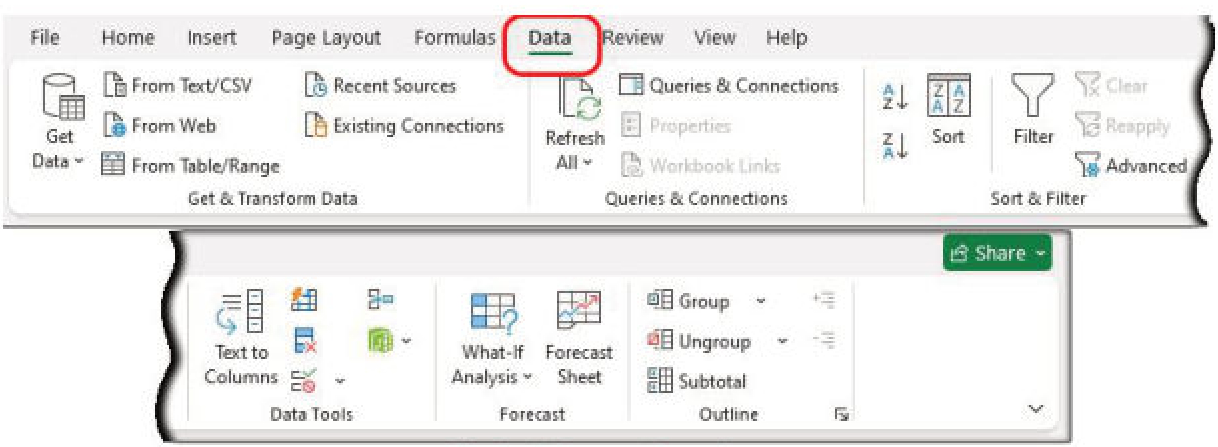


Fig 2.11 Data tab options

2.4.7 Review Tab:

The Review tab in Excel is essential for proofreading, collaborating, and protecting your workbook. Whether you are checking for spelling mistakes, adding comments, tracking changes, or securing your work.

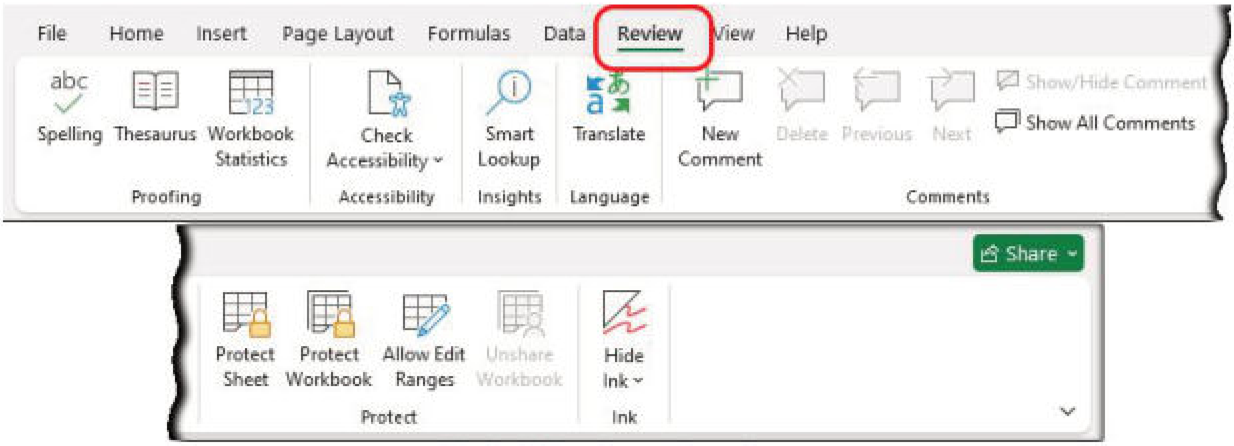


Fig 2.12 Review tab options

2.4.8 View Tab:

The View tab in Microsoft Excel help you to manage how your worksheet is displayed on your screen. It provides tools for adjusting the zoom, arranging windows, and showing or hiding elements in your workbook.

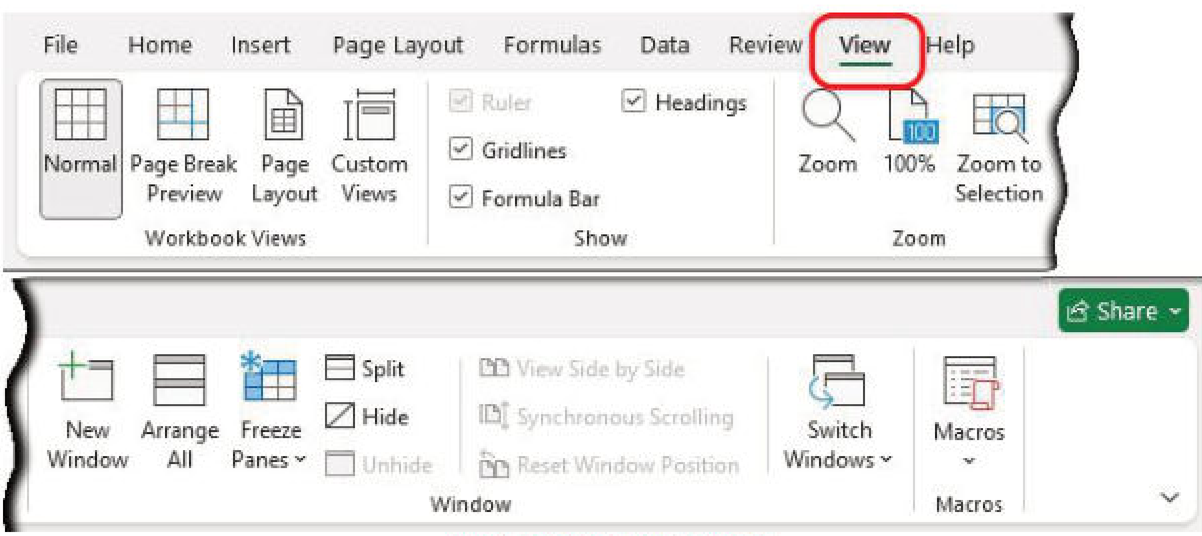


Fig 2.13 View tab options

2.5 Formulas and Functions:

In Microsoft Excel formulas and functions are essential tools that perform calculations, analyze data, and automate tasks within worksheets. While formulas are custom calculations created by the user, functions are predefined formulas that Excel provides for specific tasks.

2.5.1 Formula:

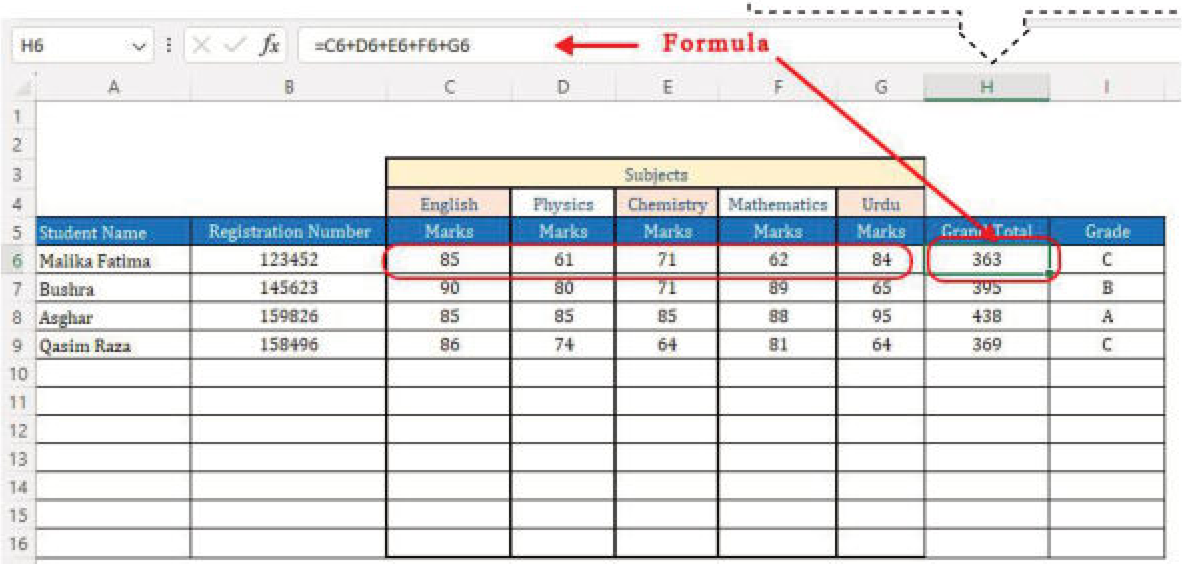
A formula is an equation that performs calculations on data in Excel worksheet. It can be simple or complex, and allow us to perform arithmetic operations, manipulate text, reference other cells, or perform logical tests.

Basic Formula Structure: A formula always starts with an equals sign (=). After the equals sign, you can include numbers, cell references, operators (like +, -, *, / etc.), and functions. Here are some examples:

Formula Expression	Description
=A1 + B1	This adds the values in cells A1 and B1
=A2 * B2	This multiplies the values in cells A2 and B2.
=A1 - B1 + C1	This subtracts B1 from A1 and then adds with the value in C1

Following is the example of student total marks:

Tip: Total marks of students can also be calculated with Auto Sum ∑



	Subjects						
		English	Physics	Chemistry	Mathematics	Urdu		
Student Name	Registration Number	Marks	Marks	Marks	Marks	Marks	Grand Total	Grade
Malika Fatima	123452	85	61	71	62	84	363	C
Bushra	145623	90	80	71	89	65	395	B
Asghar	159826	85	85	85	88	95	438	A
Qasim Raza	158496	86	74	64	81	64	369	C

Fig 2.14 (a) Basic Addition Formula

Another example to calculate the percentage of a student's marks using formula shown below:

I6 =H6/500 *100

	A	B	C	D	E	F	G	H	I	J
1										
2										
3			Subjects							
4			English	Physics	Chemistry	Mathematics	Urdu			
5	Student Name	Registration Number	Marks	Marks	Marks	Marks	Marks	Grand Total	Percentage	Grade
6	Malika Fatima	123452	85	61	71	62	84	363	72.6	A
7	Bushra	145623	33	80	50	45	34	242	48.4	D
8	Asghar	159826	85	38	85	33	37	278	55.6	C
9	Qasim Raza	158496	74	74	64	70	64	346	69.2	B
10										
11										
12										
13										
14										
15										
16										

Fig 2.14(b) Basic Percentage Formula

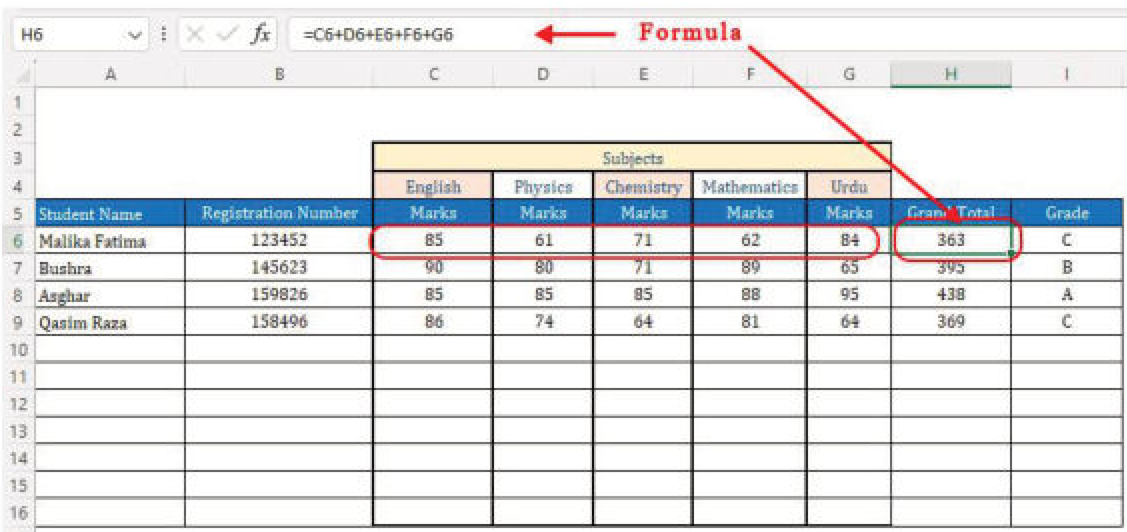
2.5.2 Function:

A function is a predefined formula in Excel that helps you perform common calculations without having to manually create a formula. Excel comes with a wide variety of functions that simplify complex tasks like summing numbers, averaging data, counting items, maximum value, minimum value and much more.

Function Structure: A function always starts with an equal sign (=) and the function name, followed by arguments enclosed in parentheses. For example:

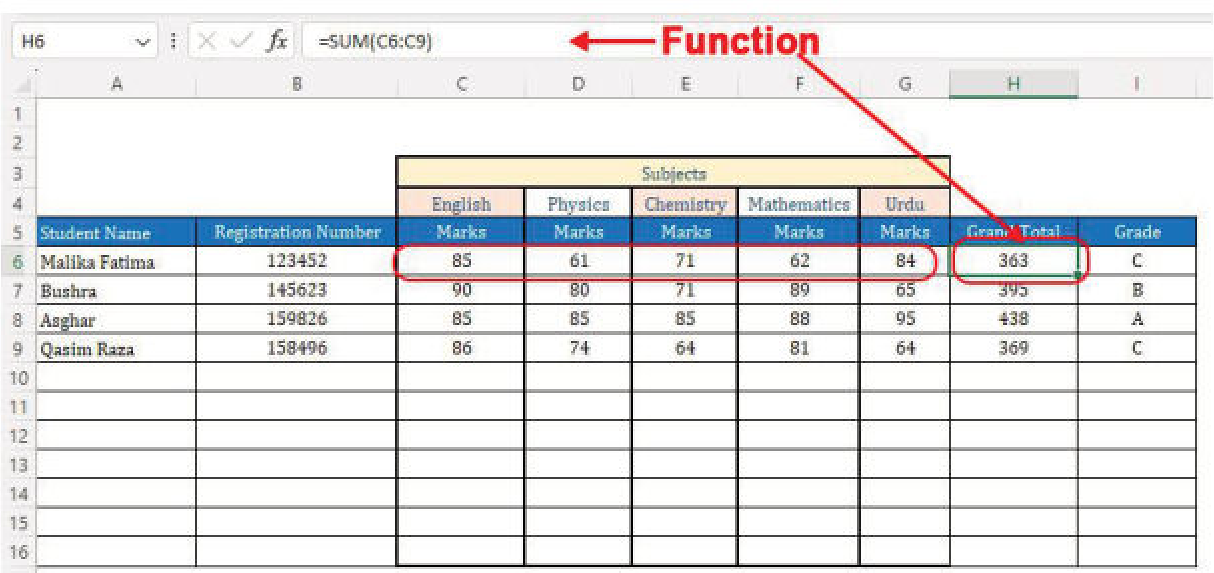
Function Expression	Description
=SUM(C6:C9)	This function adds up all the numbers in cells range C6 through C9. **Fig 2.15(a), Fig 2.15(b)** and **Fig 2.15(c)**
=AVERAGE(C6:G6)	This function calculates the average (mean) of the numbers in cells C6 to G6. **Fig 2.15(d)**
=IF(H6 > 10, "Yes", "No")	This is a logical function that checks if the value in cell A1 is greater than 10. If true, it returns "Yes"; otherwise, it returns "No." **Fig 2.15(e)**

Instead of selecting each cell for addition from C6 to C9 like **C6+C7+C8+C9** you can select range of cells by using Colon sign (**:**) e.g. **C6:C9** and use Sum function i.e. **Sum(C6:C9)**



	Student Name	Registration Number	English Marks	Physics Marks	Chemistry Marks	Mathematics Marks	Urdu Marks	Grand Total	Grade
6	Malika Fatima	123452	85	61	71	62	84	363	C
7	Bushra	145623	90	80	71	89	65	395	B
8	Asghar	159826	85	85	85	88	95	438	A
9	Qasim Raza	158496	86	74	64	81	64	369	C

Fig 2.15(a) Formula to calculate the addition of columns



	Student Name	Registration Number	English Marks	Physics Marks	Chemistry Marks	Mathematics Marks	Urdu Marks	Grand Total	Grade
6	Malika Fatima	123452	85	61	71	62	84	363	C
7	Bushra	145623	90	80	71	89	65	395	B
8	Asghar	159826	85	85	85	88	95	438	A
9	Qasim Raza	158496	86	74	64	81	64	369	C

Fig 2.15(b) Showing Sum function calculation using column range C6:C9

fx Insert Function

To see all the functions by category, choose **Formulas Tab » Insert Function**. Insert function Dialog appears from which we can choose the function.

H6 | f_x | =SUM(C6,D6,E6,F6,G6) ← **Function**

Student Name	Registration Number	English Marks	Physics Marks	Chemistry Marks	Mathematics Marks	Urdu Marks	Grand Total	Grade
Malika Fatima	123452	85	61	71	62	84	363	C
Bushra	145623	90	80	71	89	65	395	B
Asghar	159826	85	85	85	88	95	438	A
Qasim Raza	158496	86	74	64	81	64	369	C

Fig 2.15(c) Showing Sum function with different Column number

I6 | f_x | =AVERAGE(C6:G6)

Student Name	Registration Number	English Marks	Physics Marks	Chemistry Marks	Mathematics Marks	Urdu Marks	Grand Total	Average	Grade
Malika Fatima	123452	85	61	71	62	84	363	72.6	A
Bushra	145623	33	80	50	45	34	242	48.4	D
Asghar	159826	85	38	85	33	37	278	55.6	C
Qasim Raza	158496	74	74	64	70	64	346	69.2	B

Fig 2.15(d) Showing Average function with Column range

I6 | f_x | =IF(H6/500>=0.7, "A", IF(H6/500>=0.6, "B", IF(H6/500>=0.5, "C", "D")))

Student Name	Registration Number	English Marks	Physics Marks	Chemistry Marks	Mathematics Marks	Urdu Marks	Grand Total	Average	Grade
Malika Fatima	123452	85	61	71	62	84	363	72.6	A
Bushra	145623	33	80	50	45	34	242	48.4	D
Asghar	159826	85	38	85	33	37	278	55.6	C
Qasim Raza	158496	74	74	64	70	64	346	69.2	B

Fig 2.15(e) IF function to calculate the Grade

Some Examples of built-in Function:

	A	B	C	D	E
1	Numbers		Min:	1	=MIN(A2:A9)
2	1				
3	9		Max:	10	=MAX(A2:A9)
4	2				
5	5		Average:	5.75	=AVERAGE(A2:A9)
6	7				
7	4		2nd smallest:	2	=SMALL(A2:A9, 2)
8	10				
9	8		3rd largest:	8	=LARGE(A2:A9, 3)

Fig 2.15(f) Some basic built-in functions

2.5.3 Sorting Data:

Sorting is the process of arranging data into specific order like ascending or descending. Data showing in the following figure shows the data sorted percentage wise in descending order (Largest to Smallest value).

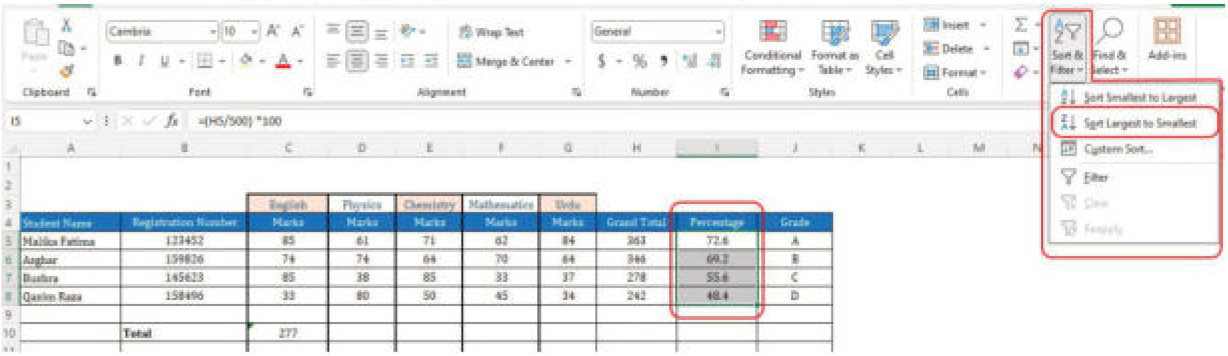


Fig 2.16 Sorting Data

Activity:

Personal Expense Calculator:

We are going to format and organize data for a simple personal expense calculator by performing following steps.

1. Open Excel software and select a blank workbook.
2. Create Headers in row 1, write the following headers starting from column A:

	A	B	C	D
1	Date	Category	Description	Amount

- **Date:** The date when the expense occurred.
- **Category:** The type of expense (e.g., Food, Rent, Transport, etc.).
- **Description:** Optional details about the expense (e.g., "Dinner at a restaurant").
- **Amount:** The cost of the expense.

3. Apply cell formatting by using Home tab (*Font group, Alignment group, Number group etc.*).
   - Apply font (bold, Time New Roman, Size etc.)
   - Apply alignment like center, left, right etc.
   - Apply borders.
   - You can increase/ decrease the cell width or height by selecting the corner of the cell and drag it as showing in the **figure 1** or right click on the column/row and select column width. A dialogue box is appeared in which you can enter the value to increase the width of column as showing in the **figure 2**.

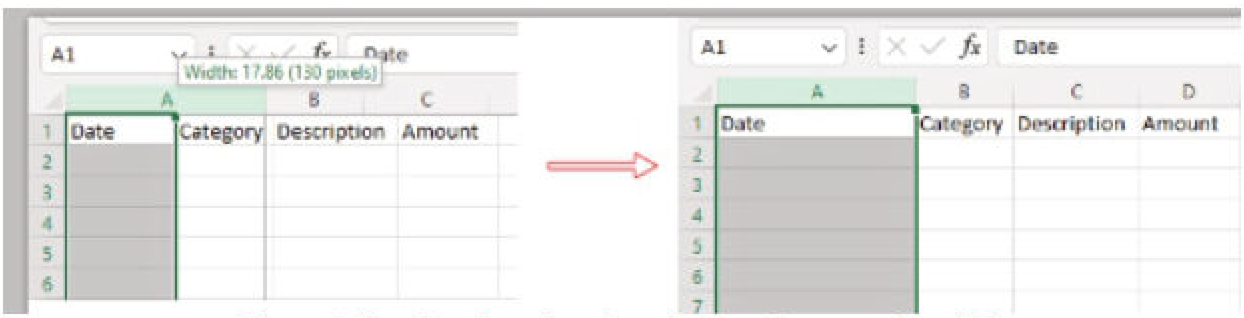


Figure 1 Showing dragging the column to increase the width

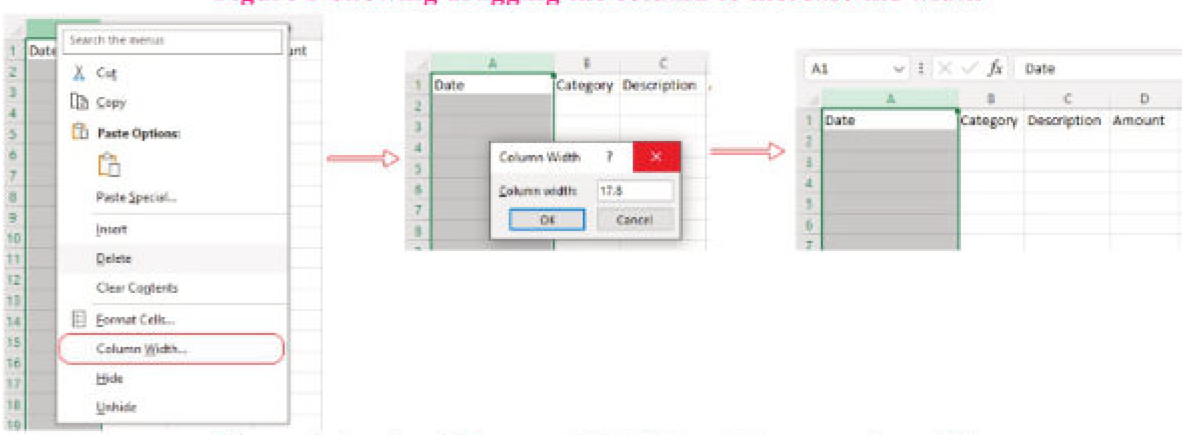


Figure 2 showing Column width dialog to increase the width

	A	B	C	D
1	Date	Category	Description	Amount
2				
3				
4				
5				
6				
7				
8				
9			Total	

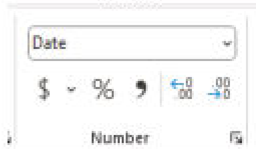
Figure 3 Table design for entering data

4. Enter some sample data in the above showing table.

Date	Category	Description	Amount
1/11/2024	Food	Grocery shopping	50
2/11/2024	Transport	Bus fare	15
3/11/2024	Rent	Monthly rent payment	500

5. Modify the Date and Amount columns' formats as indicated in the following figure. Using **Home** tab and use the **Number formatting** group to achieve it. Or use short key Ctrl +1 to format.

Hint:



Date	Category	Description	Amount
1/11/2024	Food	Grocery shopping	50
2/11/2024	Transport	Bus fare	15
3/11/2024	Rent	Monthly rent payment	500

Date	Category	Description	Amount
Monday, November 11, 2024	Food	Grocery shopping	$50.00
Tuesday, November 12, 2024	Transport	Bus fare	$15.00
Wednesday, November 13, 2024	Rent	Monthly rent payment	$500.00

6. Add total expenses in column D beside the Total cell in Row 9.
   - Use Sum() function to calculate the total amount
   - =SUM(D2:D8) - where D8 based on the number of rows in your table.

	A	B	C	D
1	Date	Category	Description	Amount
2	Monday, November 11, 2024	Food	Grocery shopping	$50.00
3	Tuesday, November 12, 2024	Transport	Bus fare	$15.00
4	Wednesday, November 13, 2024	Rent	Monthly rent payment	$500.00
5				
6				
7				
8				
9			Total	=Sum(D2:D8)

7. Categorize Expense to create the summary from the above data. Create table to list the categories as showing in following table.

	A	B	C	D
1	Date	Category	Description	Amount
2	Monday, November 11, 2024	Food	Grocery shopping	$50.00
3	Tuesday, November 12, 2024	Transport	Bus fare	$15.00
4	Wednesday, November 13, 2024	Rent	Monthly rent payment	$500.00
5	Thursday, November 14, 2024	Transport	Train Fare	$10.00
6				
7				
8				
9			Total	$575.00
10				
11				
12	Category	Total		
13	Food			
14	Transport			
15	Rent			

8. Use SUMIF() built-in function to calculate totals by Category.

Hint:

SUMIF

SUMIF(range,criteria,sum_range)

Adds the cells specified by a given condition or criteria.

SUMIF function uses range to find the criteria, if criteria successfully meets than it will sum the values from the range.

9. In the cell next to each category total use as follows
   a. for Food = SUMIF(B2:B8, "Food", D2:D8)
   b. for Transport = SUMIF(B2:B8, "Transport", D2:D8)
   c. for Rent = SUMIF(B2:B8, "Rent", D2:D8)

SUM =SUMIF(B2:B8, "Food", D2:D8)

SUMIF(range, criteria, [sum_range])

	A	B	C	D
1	Date	Category	Description	Amount
2	Monday, November 11, 2024	Food	Grocery shopping	$50.00
3	Tuesday, November 12, 2024	Transport	Bus fare	$15.00
4	Wednesday, November 13, 2024	Rent	Monthly rent payment	$500.00
5	Thursday, November 14, 2024	Transport	Train Fare	$10.00
6				
7				
8				
9			Total	$575.00
10				
11				
12	Category	Total		
13	Food	"Food", D2:D8)		
14	Transport			
15	Rent			

10. Now you expense worksheet is completed. Please save your file with the name Personal Expense.xls

	A	B	C	D
1	Date	Category	Description	Amount
2	Monday, November 11, 2024	Food	Grocery shopping	$50.00
3	Tuesday, November 12, 2024	Transport	Bus fare	$15.00
4	Wednesday, November 13, 2024	Rent	Monthly rent payment	$500.00
5	Thursday, November 14, 2024	Transport	Train Fare	$10.00
6				
7				
8				
9			Total	$575.00
10				
11				
12	Category	Total		
13	Food	50		
14	Transport	25		
15	Rent	500		

Note: You can add more functions to calculate Highest expense, Lowest expense etc. Add more conditional formatting to enhance your worksheet.

2.6 Charts:

Charts are powerful tool in Microsoft Excel that helps visually represent data, making it easier to analyze and understand patterns, trends, and relationships. Instead of displaying raw numbers, charts allow to present information in a more accessible and visually appealing way. Microsoft Excel provides range of chart types to create different types of charts e.g. column chart, bar chart, line chart, pie chart etc.

Insert Charts:

1. Select the required cells that contains data to be used in the chart. Column A (Student Name) with cell range A3:A6 and Column I (Percentage) with cell range I3:I6 are used for 3D bar chart.
2. Click on the Insert tab.

Teacher should explain the Chart details like X-axis and Y-axis, Legend, Data label, Chart Title etc. to student in computer lab.

3. Click on Chart command button.
4. Click the required type of Chart.
5. 3D bar chart is inserted as shown in the picture.

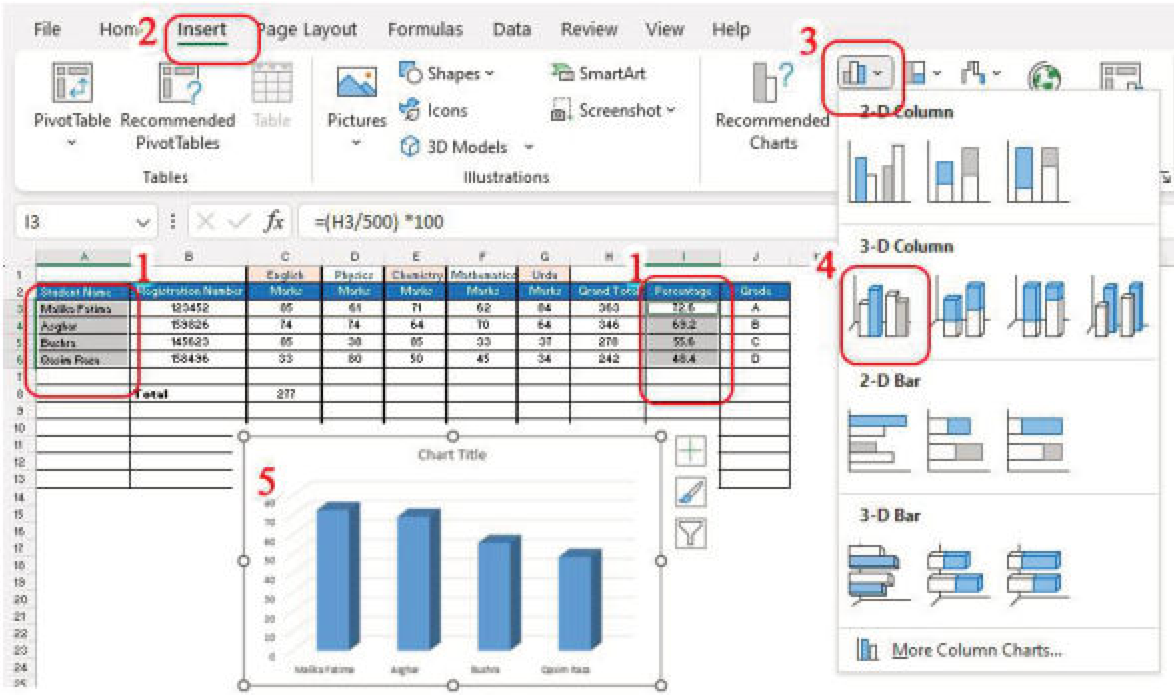


Fig 2.17 Insert Bar chart

Summary

- A **spreadsheet** software is an application software used for analysis and storage of data in tabular form.
- **Microsoft Excel** is a spreadsheet software that is used to organize and manipulate data using formulas and functions.
- **Workbook** is a collection of one or more worksheets in a single file.
- **Microsoft Excel** worksheet is a collection of cells organized in rows and columns that allows you to organize information as a table.
- **Home tab** provides groups of clipboard, editing, styles, font and paragraph; to change settings of spreadsheet such as font type and size, bullets etc.

- **Insert tab** helps to insert different objects in the spreadsheet, like tables, word art, symbols, header, footer, text boxes, equations etc.
- **Page layout** tab is used to set margins, page orientation, page size, line numbers and paragraph etc.
- **Formulas tab** provides many Excel functions categorized in different groups like math, financial, statistical, logical etc.
- **Data tab** handles huge amount of data and provides functions related to it, like importing data etc.
- **Review tab** is used to review spreadsheet for tasks like spell check, dictionary, translation etc.
- A **Formula** is an expression that operates on values in a cell or a range of cells. A formula always begins with an equality sign (=).
- A **Function** is a defined formula that performs calculation using specific values in a particular order.
- Sorting is the process of arranging data in specific order.
- Chart is graphical representation of data in a worksheet.

Workbook	Worksheet
A workbook is the entire file you create and save in Excel	A worksheet is a single page within a workbook. It is where you actually enter and organize your data, and it consists of a grid of rows and columns (cells).
A workbook can contain one or more worksheets, and each worksheet is a separate page within the workbook.	You can have many worksheets in one workbook, each with its own set of data.
It is like a binder that holds all the sheets, charts, and data.	worksheet represented by a tab at the bottom of the screen

Exercise

1. Encircle the correct answers.

i. Microsoft Excel is:
   a) Spreadsheet b) Database Management
   c) Presentation d) Workbook

ii. The intersection of a column and a row in a worksheet is called____
   a) Column b) Value
   c) Cell d) Address

iii. Excel formulas start with?
   a) / b) [ ] c) = d) &

iv. Microsoft Excel feature lets you show only the data in a column (or columns) based on certain criteria?
   a) Sorting b) Formula
   c) Filtering d) Pivot

v. In Excel, Conditional formatting is in: _______________?
   a) Insert tab b) Page layout tab
   c) Data tab d) Home tab

vi. Microsoft Excel function that checks whether a condition is true or false?
   a) SUM() b) IF()
   c) MIN() d) AVERAGE()

2. Provide the descriptive answers of the following questions.

i. Write down the uses of Microsoft Excel?
ii. What is the purpose of formula bar?
iii. Differentiate between workbook and worksheet.
iv. How to add chart into worksheet. Write down the steps.

Important points to remember

- **Spreadsheet:** Software used to store, organize, and analyse data.
- **Tabular form:** Data representation in rows and column
- **Cell:** The basic unit of a worksheet where you can enter data. It is identified by a combination of column letter and row number (e.g. A1, B2)
- **Row:** A horizontal set of cells in a worksheet identified by a number.
- **Column:** A vertical set of cells in a worksheet. It is identified by a letter.
- **Range:** A selection of two or more cells. Represented by the starting and ending cells (e.g., A1:A10).
- **Chart:** A visual representation of data, such as bar, line, or pie charts.
- **Filter:** A tool to display only the rows that meet certain criteria.
- **Ribbon:** The toolbar at the top of the Excel window containing tabs and commands.
- **Cell format:** The way a cell's content is displayed, such as number, text, currency, or date.
- **Clipboard:** A temporary storage area for cut or copied data.
- **Auto fill:** A feature that fills data automatically based on a pattern or sequence.
  - Example: Drag a cell with "1" down to fill cells with "2," "3," "4," etc.

Instruction for Teachers:

- Teachers should motivate students to explore various resources and continue learning independently.
- Divide the class into group of 2 or 3 students.
- Showing all the commands from the each Ribbon of Microsoft Excel.

i. Consider the following worksheet containing the budget and gross of each movie.

Movie	Budget ($)	World Gross ($)	Profit
Spider-Man 3	258,000,000	887,436,184	
King Kong (2005)	207,000,000	553,080,025	
Superman Returns	204,000,000	391,081,192	
Spider-Man 2	200,000,000	784,024,485	
Chronicles of Narnia, The	180,000,000	748,806,957	
Terminator 3: Rise of the Machines	170,000,000	433,058,296	
Shrek the Third	160,000,000	733,012,359	
Harry Potter and the Goblet of Fire	150,000,000	892,213,036	
Harry Potter and the Order of the Phoenix	150,000,000	822,828,538	
Mission: Impossible III	150,000,000	397,501,348	
Troy	150,000,000	497,298,577	

Highest	
Lowest	
Average	

- ➢ Write the formula to calculate the profit.
- ➢ Write the formula to find out the highest profit value.
- ➢ Write the formula to find the lowest profit value.
- ➢ Write the formula to find the average profit value.
- ➢ Sorting the data from highest profit to lowest profit movie.
- ➢ Draw pie chart on the data to show the world gross of each movie.

ii. Prepare the following worksheets and also draw the charts.

- o Student Marksheet
- o Attendance Register
- o Electricity bill

	A	B	C	D	E	F
1	Toys Ordered	Price Each	Quantity Ordered	Cost	Discount	Final Cost
2	BBQ Barbie Doll	12.99	2			
3	Price Eric Dol	8.99	3			
4	Princess Jasmine Doll	9.99	1			
5	Cinderella coach	19.99	1			
6	Spiderman gloves	14.99	3			
7	Total			-	-	-
8						

➢ In cell D2 create a formula to calculate the Cost using the following information:
Cost = Price Each * Quantity Ordered

- Copy the formula down the column to work out the other costs.
- In cell E2 create a formula to calculate the discount using the following information:
  Discount = 7 % of Cost (= 7% * Cost cell)
- Copy the formula down the column to work out the other discounts.
- In cell F2 create a formula to calculate the Final Cost using the following information:
  Final Cost = Cost - discount
- Copy the formula down the column to work out the other final costs.
- In cell B7, create a formula that will add together all the Price Each figure.
- Copy the total formula across the row to calculate the other column totals.

SOME COMMON SHORTCUT KEYS	
Ctrl + N	Opens new / blank workbook
Ctrl + A	Select all contents of the document
Ctrl + B	Bold highlighted selection
Ctrl + C	Copy selected object
Ctrl + X	Cut / move selected object
Ctrl + V	Paste, cut/copied object
Ctrl + S	Save the document
Ctrl + P	Open the print window
Ctrl + Z	Undo last action
Ctrl + Y	Redo the last action performed
Shift + F3	Insert function
F1	Help
F7	Check spelling
F11	Insert new chart
Shift + F7	Thesaurus
Shift + F11	Insert new worksheet
Alt + F4	Exit workbook
Alt +=	Autosum